A fantastic title about minihalos that somehow grips a viewers attention for over 100 pages of detail

by

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Abstract

A truly brilliant abstract that captures the nuance of what I have done, without being overly detailed and dense so that a reader throws it out the window.

Acknowledgments

People who helped me during my Thesis: Yvette, CASA people, sibling.

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Chapter 1

Introduction

1.1 Galaxy Clusters

- What is a Galaxy Cluster (grav bound galaxies, in a bunch of hot gas)
- Generic information about galaxy clusters (yay statistics)
- Fun facts about galaxy clusters

1.2 Observational work with Galaxy Clusters

- Well, we can't just go there so we get to look at them really closely.
- How do we get data about clusters
- In increasing wave length, Optical, X-ray, Radio, Gamma.
- Some telescope + research that we can do with each of these.

1.3 Radio Emission

- This is what we are interested in, can gain information about clusters from this.
- Link synchrotron emission to the electron / gas properies of the cluster and how radio emission can inform us of what the cluster is doing.
- Common sources of relativistic electrons are the AGNs, lose energy over time.
- Structures larger than the lifetime of these jets, more diffuse (Halos).

1.4 Minihalos

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- What is a minihalo (diffuse radio emission in the core of a cluster.)
- They are somewhat common, things about detections.
- Difficulty in distinguishing between production mechanisms.

1.5 Our goals

- We want to be able to measure minihalo flux for this subset of minihalos at high radio frequencies so that we can investigate if steepening is occuring.
- This is the layout of the paper and what each chapter will cover in pursuit of said goal.

Chapter 2

Background

This chapter would be about radio interferometry, how it works to some extent and the data that is retrieved from it. Then we follow up with more specific data about radio emission.

2.1 What is an interferometer

- Limits of dish size in resolution, causes us to explore different ideas.
- How does an interferometer work.
- Bunch of dishes being interfered, delay to tune positioning, uv plane to determine coverage.
- From all of this we retrieve amp + phase measurements.

2.2 Radio emission in more detail

2.2.1 Large Radio Halos

- Use similar physics but much larger scale.
- Some basics things about previous research in differentiating production.
- Gamma ray non-detection.
- General consensus is that its re-acceleration.

2.2.2 Minihalos

- These are much smaller, so therefore much harder to observe. Difference to large radio halos.

- Similarities between host clusters (CC), general properties of spectral index etc.

2.2.3 Hadronic Production

- What is the process for hadronic production
- How does this manifest spectrally and evolve. What should we expect observationally here based on simulation work.
- How does this manifest spatially and evolve, what should we see observationally.

2.2.4 Turbulent re-acceleration

- What is the process here.
- How manifest spectrally.
- How manifest spatially.

2.2.5 Previous minihalo studies

- Generally look at lower frequencies, or a singular study for a particular minihalo.
- Want to look at more minihalos in a more holistic manner.
- Some findings from previous papers e.g. RXJ1720, MS1455+2232 in basic detail. (more detail on the per-halo discussion).

2.3 Our observations

- We are using VLA data in X-band, what does this mean and what are the specs for the VLA.
- These observations are of list minihalos and this setup, observation proposal.
- Original source of these minihalos is the G14 paper (among others).
- Table of minihalos + observing setup.

Chapter 3

Calibration and Imaging

This would be how to calibrate and the calibration results for the datasets, plus any special considerations we took.

3.1 How to calibrate a dataset

- We have the VLA calibration pipeline to remove stress from our lives.
- We have a non-neglible amount of RFI to add stress to our lives.
- A variety of steps but first pain, what is RFI and what does it cause.

3.2 Flagging data

- Initial testing showed that the pipeline flagging steps didn't work well for our data.
- AOFLAGGER worked much better and has a mostly out-of-box VLA script.
- How does AOFLAGGER work
- Integrating into a pipeline step to still use the pipeline

3.3 Calibration pipeline

- Work through each calibration step explaining what it does and why it is important.
- Final preparation for imaging is averaging down to further reduce noise.
- Notes about general trends from the calibration, with RFI in 10-12 GHz D config being bad, much of it being flagged.

3.4 Special mentions

- Either here or during following imaging section talk about the 2-3 datasets we processed slightly differently due to bad fits.
- Describe the fitting process for the custom bandpass.

3.5 Imaging

- When we make a raw image from the data, we still have a lot of artefacts from the point spread function, which comes from the beam.
- This means we need a more advanced imaging process to remove this to get a clean image. Implemented in a few different ways, we choose tclean in CASA.
- Tclean works by modelling the PSF, creating an empty model, find brightest point in residuals, model that as real, subtract model from residuals, repeat 20K times.
- This works mostly well, but we can improve results using self calibration.

3.5.1 Self calibration

- The phase calibration is the most iffy part for the main calibration, so we can make a test image and check the model seems reasonable.
- Then use that model to make small phase corrections to our dataset. Then repeat that a few times and these corrections should converge since it is circular.
- This is the final calibration steps that we do, phase corrections 20 degrees $\rightarrow$ 1 degree after a few rounds.

3.5.2 Final cleaning

- Between self calibrations this we also check the datasets and do some final manual cleaning.
- Clipping the datasets to a sensible amp range, removing spikes, SPWs where there is small amounts of data left.
- Overview of the `assess_spw` function used for this.

Chapter 4

How to Analysis

This chapter starts with an overview on imaging and how that is performed. Then we talk a bit about how we actually want to measure the flux of the minihalos. Could combine with actual results.

4.1 Final datasets

- Could have a table here talking about each minihalo, how much data is remaining, spws that are flagged out, some sort of visual.
- Maybe bar graph of spws, line or point graph for minihalo C/D (either total or differentiated by style) normalised to [0, 1]

4.2 Calibration Error

- Calibration error exists, and sources can vary in their flux amount.
- We can try and constrain the error and variance by comparing the secondary calibrator models for D config and the point source fluxes for the images.
- Fluxes for secondaries can vary a lot over a year (100%?) maybe, but lower per month. Can then compare to differences in sources in the image.
- If a trend is observed, likely calibration error. If no trend or close models, constrain calibration error and can see the variance in AGN etc.
- Can retrieve the point source flux by cutting out diffuse sources (uvrange) and by masking to a tight border.

4.3 Calculating the minihalo flux

- We can do this in a variety of ways, but we don't want the AGN contamination so we should subtract this out.
- Do this per dataset to account for variation over time, incorrect subtractions can result in negative bowls in emission, or remaining flux.
- Other problematic sources are dealt with the same way.
- Image the subtracted data and then we create a contour for significant emission (3 sigma) from the residual RMS.
- Calculate total flux in the region.
- Then we calculate the error using this formula from G14 modified for the AGN error since we think its lower.

Chapter 5

Results

- Sections for each minihalo with a brief explanation of imaging process, custom things we had to do to get results.
- Number of self calibration, sources subtracted, difficulties in imaging certain aspects.
- Discussion on the calibration error and variance in the observed point sources.
- Includes an image of the uvrange data showing point sources, image of the minihalo with contours, and anything else we desire.
- Includes discussion compared to previous results or measured values (likely G14) and any caveats or concerns.

Chapter 6

Discussion

- General findings from the above
- Any downsides caveats for the whole dataset
- Compare general results to things like G14 / G19 and how comparable these values are. Both literally and if valid comparison.

Chapter 7

Conclusion

- I'm sure I will have some conclusions here.

Bibliography